

Lab 3.3: TaskBoard: Static Task List UI

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****Goal**** | Render the TaskBoard board from mock data using a clean component tree.

****Suggested time**** | 45 min

This is your capstone for today. Notice that done has changed to status with three values, so the board can have three columns. Update the data, and any component that still reads done: Board now filters on status, and TaskCard no longer needs a Done or Open label because the column shows it. Use string ids such as "1"; they match the ids crypto.randomUUID() creates tomorrow and the Task type on Day 6.

Keep App.jsx tiny. All logic belongs in the components.

Steps

1. Put 8 sample tasks in src/data/tasks.js.
2. Give each task an id, title, assignee, points and status ("todo", "doing" or "done"). Leave assignee out of at least one task.
3. Create src/components/Header.jsx showing the app name and the total task count.
4. Update Board.jsx so it filters tasks into three Columns by status.
5. Make sure Column.jsx maps tasks to TaskCards with keys.
6. Update TaskCard.jsx to show the title, assignee and points, hiding the assignee line when it is missing.
7. Show Nothing here yet in an empty column.
8. Style the board as three columns in App.css, then commit and push:

```
bash
git add .
git commit -m "Build static TaskBoard UI"
git push
```

Done when

- Three columns show the correct tasks
- The counts in the column headings are correct
- `App.jsx` renders only `Header` and `Board`

Hints (try on your own first)

Data shape (src/data/tasks.js; add five more tasks of your own)

```
export const tasks = [
  { id: "1", title: "Set up project", assignee: "Zanele", points: 2, status: "done" },
  { id: "2", title: "Build task list", assignee: "Sipho", points: 5, status: "doing" },
  { id: "3", title: "Write README", points: 1, status: "todo" },
];
```

Board: describe the columns once as data, then map over them and filter the tasks for each status. This avoids repeating three Column lines. Because this is a list, each Column needs a

key.

src/components/Board.jsx

```
import Column from "../Column.jsx";

const columns = [
  ["todo", "To do"],
  ["doing", "In progress"],
  ["done", "Done"]
];

function Board({ tasks }) {
  return (
    <div className="board">
      {columns.map(([{status, heading}]) => (
        <Column
          key={status}
          heading={heading}
          tasks={tasks.filter((t) => t.status === status)}
        />
      ))}
    </div>
  );
}

export default Board;
```

src/App.css

```
.board {
  display: grid;
  grid-template-columns: repeat(3, 1fr);
  gap: 16px;
}
```

Troubleshooting

Problem | Fix

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Tasks appear in the wrong column, or in no column | The status spelling differs between the data and the filter, for example `"in-progress"` against `"doing"`.

Everything is in one column | The CSS for `.board` is missing `display: grid` and `grid-template-columns: repeat(3, 1fr)`.

Header count is wrong | You passed a filtered list to `Header` instead of the full list.

Every card shows "Open", or a column is empty | A component still reads `done`, which no longer exists in the data. Use `status`.

> Stretch challenge

> - Add a points total to each column heading using `reduce`.

> - Add a colour per status using a class name, for example `column column-doing`.

When you finish, open React DevTools and look at your component tree: App, Header, Board, three Columns and the TaskCards inside them.

